



Effect of color information on the diagnostic performance of glaucoma in deep learning using few fundus images

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Abstract

Purpose The purpose of this study was to evaluate the accuracy of the convolutional neural network (CNN) model in glaucoma identification with three primary colors (red, green, blue; RGB) and split color channels using fundus photographs with a small sample size.

Methods The dataset was prepared using color fundus photographs captured with a fundus camera (VX-10i, Kowa Co., Ltd., Tokyo, Japan). The training dataset consisted of 200 images, and the validation

dataset contained 60 images. In the preprocessing stage, the color channels for the fundus images were separated into red (red model), green (green model), and blue (blue model) using OpenCV on Windows. All images were resized to squares with a size of 512×512 pixels for preprocessing before input into the model, and the model was fine-tuned with VGG16. **Results** The diagnostic performance was significantly higher in the green model [area under the curve (AUC) 0.946; 95% confidence interval (CI) 0.851–0.982] than in the RGB model (AUC 0.800; 95% CI 0.658–0.893; $P = 0.006$), red model (AUC 0.746; 95% CI 0.601–0.851; $P = 0.002$), and blue model (AUC 0.558; 95% CI 0.405–0.700; $P < 0.001$). **Conclusion** The present study showed that the green digital filter is useful for structuring CNN models for automatic discrimination of glaucoma using fundus photographs with a small sample size. The present findings suggest that preprocessing, when creating the CNN model, is an important step for the identification of a large number of retinal diseases using color fundus photographs.

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Keywords Glaucoma · Fundus photography · Deep learning · Convolutional neural network

Introduction

Glaucoma is an irreversible disease and is the second most common cause of blindness worldwide [1]. A recent meta-analysis based on 50 glaucoma prevalence-related studies reported that the prevalence of glaucoma was 3.5% among individuals aged 40–80 years, 3.0% for primary open-angle glaucoma, and 0.5% for primary angle closure glaucoma, corresponding to an estimated 64.3 million individuals with the disease worldwide [2]. The population of patients with glaucoma is expected to reach 111.8 million by 2040 [2].

Early detection of glaucoma is essential to prevent vision loss. In glaucoma, typical fundus patterns develop in the optic disk. Ophthalmologists diagnose glaucoma based on the structural changes in the optic nerve head (ONH) [3] and the retinal nerve fiber layer (RNFL) [4]. The structure around the optic disk can be captured using two-dimensional fundus photography [1, 5], optical coherence tomography (OCT) [1, 6], and scanning laser polarimetry [7–9].

Two-dimensional fundus photography is the primary ophthalmological screening method globally and provides important information to prevent vision loss through early glaucoma detection [5]. Some medical check-up clinics have adopted the routine use of color fundus photographs. However, for ophthalmologists, processing a huge fundus image is a burden. Further, the diagnosis of glaucoma using fundus photographs is subjective, and the accuracy depends on the skill of the ophthalmologist [10].

Convolutional neural network (CNN), a deep learning algorithm, has recently become one of the main techniques for image recognition [11]. Recent studies have suggested that the CNN model can automatically identify glaucoma using two-dimensional fundus photographs [12–14], and the model has a slightly higher diagnostic performance compared with the human eye [10, 15]. These studies suggested that CNN is a useful tool for identifying ocular diseases, and it is anticipated that its use will reduce the burden on ophthalmologists.

The application of deep learning to medical data can be problematic, because a single hospital is incapable of accumulating a sufficiently large disease dataset. Data augmentation is a common technique to improve the performance of the CNN model [11, 16], even if the dataset is small. However, the effect of data

augmentation depends on the size of the dataset [16, 17]. The diagnostic performance in a small dataset reaches saturation faster than in a large dataset, because the small dataset has insufficient feature space and data space for data mining [17].

Monochromatic red-free fundus images using a green contrast filter are able to highlight nerve fiber layer defects in glaucoma eyes [18]. A CNN usually captures the information of the three primary colors (red, green, blue; RGB) to generate an optimal network using the color images [12, 13, 19]. We hypothesized that changing the color information of the fundus photograph during preprocessing could improve the performance of the CNN model for glaucoma to be as good as the human eye.

The purpose of this study was to compare the performance of the CNN model that reached the upper limit of performance by adjusting the parameters and the CNN model trained by changing the color information during the preprocessing using fundus photographs with a small sample size.

Methods

Subjects

Dataset

The training and validation datasets were prepared using color fundus photographs captured with a fundus camera (VX-10i using the CCD of KD-211C, Kowa Co., Ltd., Tokyo, Japan) at Inoue Eye Hospital. All fundus color photographs were taken with a pupillary diameter of equal or greater than 6.0 mm, flashlight brightness of 6.3, 9.5, or 12.5 mW, angle of view of 50°, resolution of 1604 × 1216 pixels, and jpeg compression. A total of 260 fundus images were captured from 130 patients with glaucoma (100 training and 30 validation images) and 130 healthy individuals (100 training and 30 validation images). Complete ophthalmological examinations were performed for these patients and included assessing the ONH and macula using OCT (CIRRUS HD-OCT, Carl Zeiss Meditec, Dublin, CA), intraocular pressure (IOP) using Goldmann applanation tonometer, visual field using the Humphrey perimeter (Carl Zeiss Meditec), and best-corrected visual acuity using the decimal visual acuity chart at Inoue Eye Hospital.

The test dataset was accumulated after the CNN model was completed. The color fundus photographs captured with the same fundus camera (VX-10); however, the CCD was changed (Nikon D200; Nikon Corp., Tokyo, Japan). All fundus color photographs were taken with an angle of view of 50° , resolution of 3872×2592 pixels, and jpeg compression. Then, the resolution of these images was resized to 1604×1216 pixels. Seventy-seven fundus images were captured from patients with glaucoma through complete ophthalmological examinations. Sixty-two of these patients underwent Humphrey perimetry on the same day.

All fundus images were reviewed by a single ophthalmologist who is a glaucoma specialist (K.I.). Photographs that were out of focus, unclear, too dark, too bright, decentered from the macular region, missing the optic disk, and showed other retinal diseases were excluded. The inclusion criteria for glaucoma complied with the Japan Glaucoma Society Guidelines for Glaucoma [20].

Convolutional neural network model

We fine-tuned the VGG16 model to detect glaucoma using fundus photographs [21, 22]. The CNN architecture VGG16 was utilized in the present study with weights pre-trained on the ImageNet data set for fundus feature extraction using TensorFlow 1.15.0 as a backend for Keras 2.1.6 on Python 3.6.5 for Windows. The network in VGG16 was frozen until block 4 to utilize the ImageNet weight parameters.

A kernel size of 3×3 filters with a strider of 1 was applied to the input for all convolutional layers. The flatten layer was set after block 5 of the VGG16 network. The fully connected layer (FC) after the block 5 flatten layer was newly created. FC1 had 256 units and was activated using the ReLU function and then a dropout layer with a ratio of 50%. FC2 provided normalized glaucoma data output using the softmax function. We defined glaucoma as a predicted value of ≤ 0.5 .

The hyper parameters were as follows: the activator was ReLU, the optimizer was Adam with a learning rate of $1e-5$, and the Epochs were 20 for all CNN models.

Experiment 1

In Experiment 1, we verified the diagnostic performance of the CNN model using data augmentation. The 200-image training dataset was increased to 400, 800, 1000, 1500, and 3000 images using ImageDataGenerator in Keras. The original images were randomly changed using a rotation of $\pm 5^\circ$, horizontal and vertical shift of 0–153 pixels, brightness of 0.2–1.0, and horizontal flip of 0.5. All input images were resized to squares of 512×512 pixels before input into the CNN.

Statistical analysis

To estimate the diagnostic performance of the CNN models for glaucoma, we used receiver operating characteristic curves and calculated the corresponding area under each curve (AUC), which is mathematically equivalent to the Wilcoxon signed-rank test. We compared the AUCs of the CNN models with the 200 (no-data augmentation), 400, 800, 1000, 1500, and 3000 images in the validation data. JMP Pro 14 (SAS Institute Inc., Cary, NC) was used to determine the significance of differences between the groups, and Bonferroni-corrected P value of < 0.05 was considered statistically significant.

Result 1

The diagnostic performance of the CNN model in the 1000 (AUC 0.782; 95% confidence interval (CI) 0.665–0.866; $P = 0.014$), 1500 (AUC 0.792; 95% CI 0.665–0.871; $P = 0.006$), and 3000 (AUC 0.800; 95% CI 0.658–0.893; $P = 0.003$) trained images were significantly higher than the no-data augmentation model. The diagnostic performance of the CNN model was significantly higher in the 3000 trained images than in the 400 trained images (AUC 0.712; 95% CI 0.585–0.814; $P = 0.006$; $P = 0.035$; Fig. 1). These findings suggested that the 3000 image was the upper limit of data augmentation in the trained dataset with 200 images.

Experiment 2

Experiment 2 verified the diagnostic performance of the CNN model that reached the upper limit by tuning the parameters that were changed during the

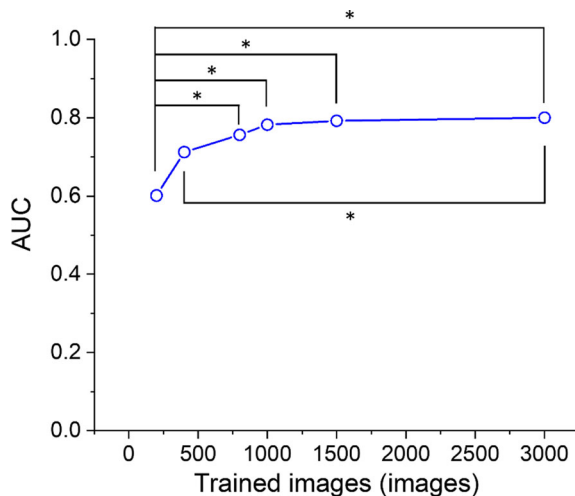


Fig. 1 Diagnostic performance depends on data augmentation. The diagnostic performance significantly improved by data augmentation. The effect of data augmentation started to saturate around 1000 images. * $P < 0.05$

preprocessing of the color information. The 3000 images used in Experiment 1 were converted to 8-bit grayscale using OpenCV 3.4.1. All input images were resized to squares of 512×512 pixels before input into the CNN.

Statistical analysis

To estimate the diagnostic performance of the CNN models for glaucoma, the AUCs of the CNN models with the 3000 grayscale and 3000 color images in the validation data of the color fundus images were determined.

Result 2

The diagnostic performance of the CNN model was not significantly different between the 3000 trained grayscale (AUC 0.851; 95% CI 0.747–0.916; $P = 0.191$) and the 3000 trained color images (Fig. 2). However, the diagnostic performance of the CNN model in the 3000 trained grayscale images improved in comparison with that in the 3000 trained color images.

These findings suggested that the change in color information during preprocessing contributed to the diagnostic performance of the CNN model for glaucoma. We considered that the grayscale algorithm was relevant to improve the diagnostic performance of the

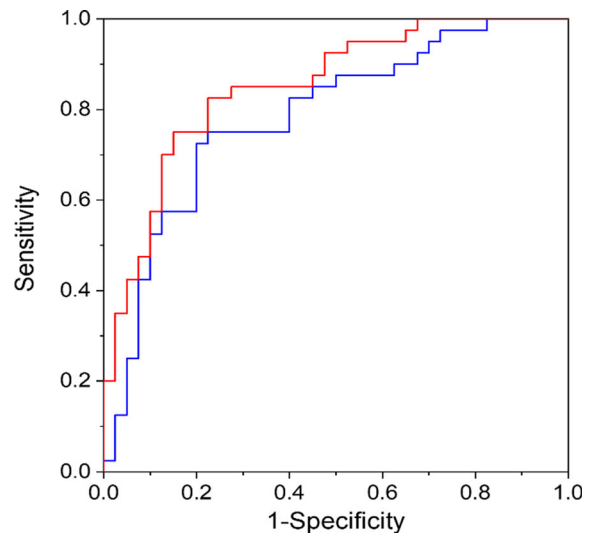


Fig. 2 Effect of grayscale. The red and blue lines indicate the diagnostic performance of the CNN model that trained 3000 grayscale and color images

CNN model. The default grayscale algorithm in OpenCV used BT.601 that was calculated in 8-bit grayscale as per the following equation: $v = 0.299 * \text{red} + 0.587 * \text{green} + 0.114 * \text{blue}$.

The result of Experiment 2 suggested that the diagnostic performance of the CNN model in Experiment 1 might be saturated by the red color information, because the fundus image has a lot of red components due to the presence of blood vessels [23]. This finding led us to speculate that the diagnostic accuracy would increase if the color information were separated.

Experiment 3

Experiment 3 verified the diagnostic performance of the CNN model that the three-color channels (red, green, and blue) in the 3000 images that used Experimental 1 were split into red (red model), green (green model), and blue (blue model) individually using OpenCV on Windows (Fig. 3). Each single-color intensity image was converted to an 8-bit grayscale image, and the file size of the single-color image decreased to $6.1\% \pm 2.9\%$ compared to the color image. All input images were resized to squares with a size of 512×512 pixels as preprocessing before input into CNN. The time per epoch was calculated by datetime module.

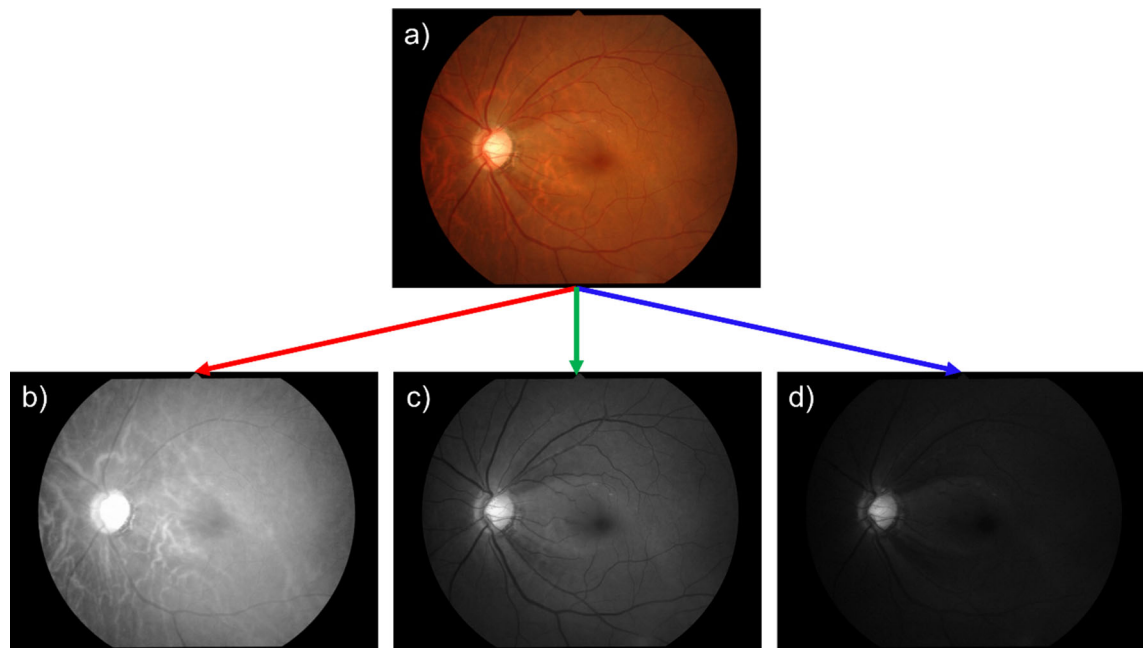


Fig. 3 RGB (a) and splitting to red (b), green (c), and blue (d) channels in the fundus photograph. In the preprocessing stage, the three-color channel of the fundus images was separated into red, green, and blue individually using OpenCV on Windows

Visual explanation

Gradient-weight class activation mapping (Grad-CAM) is one of the approaches used to visualize the important regions of an input image predicted by the CNN model [24]. Grad-CAM can generate visual explanations from any CNN-based network without requiring architectural changes or re-training. Grad-CAM images were generated by the feature map in the last convolutional layer.

Statistical analysis

To estimate the diagnostic performance of the CNN models for glaucoma, we determined the AUCs of the RGB, red, green, and blue models in the validation data for the color fundus images. The time per epoch in each model was assessed using Dunn test.

The relationships between the logarithmic converted predictions (log-predictions) of the test data in all CNN models and mean deviation (MD) were assessed using single linear regression analysis.

JMP Pro 14 (SAS Institute Inc., Cary, NC) was used to determine the significance of differences, and a Bonferroni-corrected P value less than 0.05 was considered statistically significant.

Result 3

The diagnostic performance of the validation data was significantly higher in the green model (AUC 0.946; 95% confidence interval (CI) 0.851–0.982) than in the RGB model (AUC 0.800; 95% CI 0.658–0.893; $P = 0.006$), red model (AUC 0.746; 95% CI 0.601–0.851; $P = 0.002$), and blue model (AUC 0.558; 95% CI 0.405–0.700; $P < 0.001$; Fig. 4). Furthermore, the diagnostic performance of the validation data was significantly higher in the RGB model than in the blue model ($P = 0.005$; Fig. 4). The time per epoch was significantly faster in the red (212.20 ± 7.71 s), green (210.19 ± 6.17 s), and blue (208.55 ± 6.29 s) models than in the RGB model (230.75 ± 6.76 s; $P < 0.001$).

The diagnostic performance of the test data in the green model (0.974) was the highest among the four models (RGB model 0.316; red model 0.456; blue model 0.038; Fig. 5). The representative visual explanations for glaucoma in each model are shown in Fig. 6. The green model had the highest diagnostic performance among the four models focused to the optic disk and RNFL in the fundus photograph (Fig. 6c, g). Furthermore, log-predictions in the green model were significantly and positively correlated

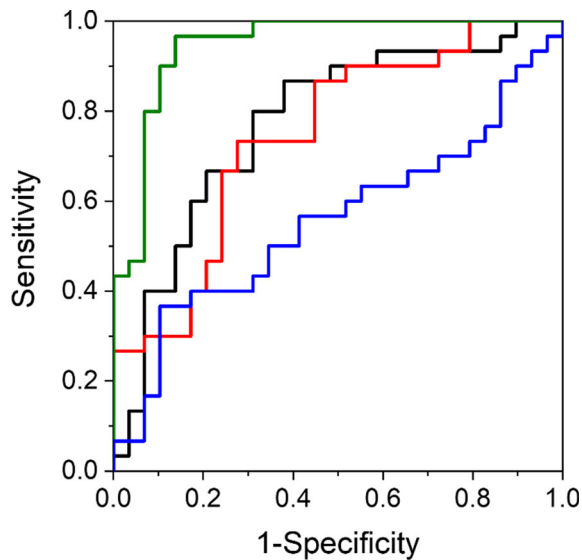
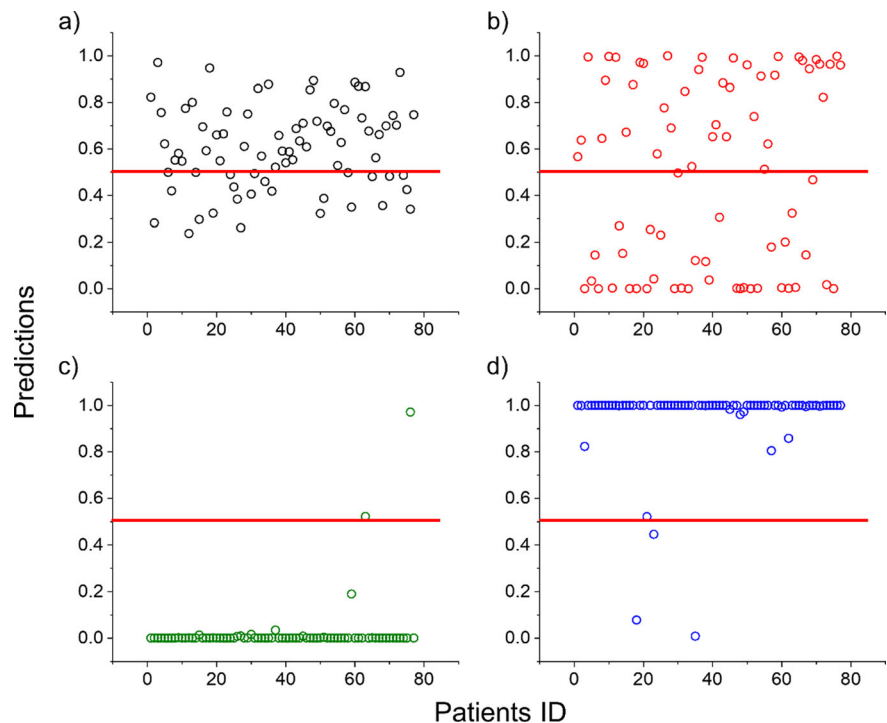


Fig. 4 Diagnostic performance in RGB, red, green, and blue CNN models. The black, red, green, and blue lines indicate the diagnostic performance of the RGB, red, green, and blue CNN models, respectively. AUC was significantly higher in the green model than in the RGB, red, and blue models. *CNN* convolutional neural network, *AUC* area under the curve

with those of MD (adjusted $R^2 = 0.137$, $P = 0.002$; Fig. 7c).

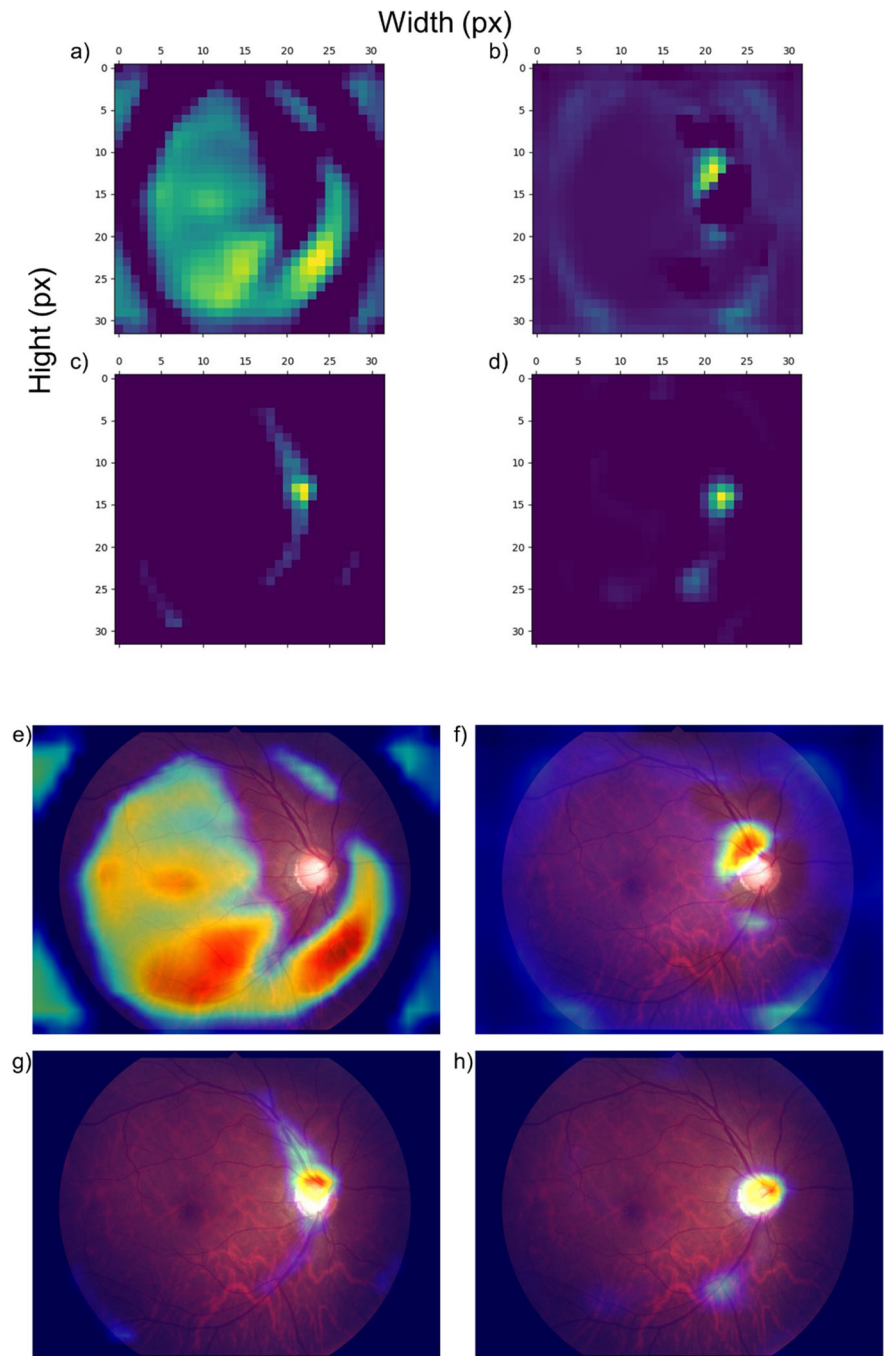
Fig. 5 Diagnostic performance in test dataset using different fundus cameras. The black, red, green, and blue circles indicate the diagnostic performance of the RGB (a 0.316), red (b 0.468), green (c 0.974), and blue (d 0.038) models, respectively



Discussion

In the present study, we evaluated improvement in the diagnostic performance of the CNN model for glaucoma that reached the upper limit by tuning the parameters used by changing and splitting of color channels in the preprocessing of a small sample size of fundus photographs. The diagnostic performance for glaucoma that reached the upper limit by tuning the parameters slightly improved by converting it to grayscale (Fig. 2). In clinical practice, a green contrast filter is used with monochromatic red-free fundus images to highlight nerve fiber layer defects [18]. The color fundus image has a lot of red components due to blood vessels, providing information pertaining to the RNFL [23, 25]. The diagnostic performance for glaucoma was significantly higher in the green model than in the RGB model, and the training time was also significantly shorter in the green model than in the RGB model in Experiment 3 (Figs. 4, 5). The present findings are in agreement with those reported by Lee et al. [26]. Lee et al. used Inception-v3 (Google Inc.) with red-free fundus photos, and their diagnostic performance for glaucoma was 0.939, similar to that of our green model (validation accuracy, 0.946; test accuracy, 0.974). Furthermore, the green model

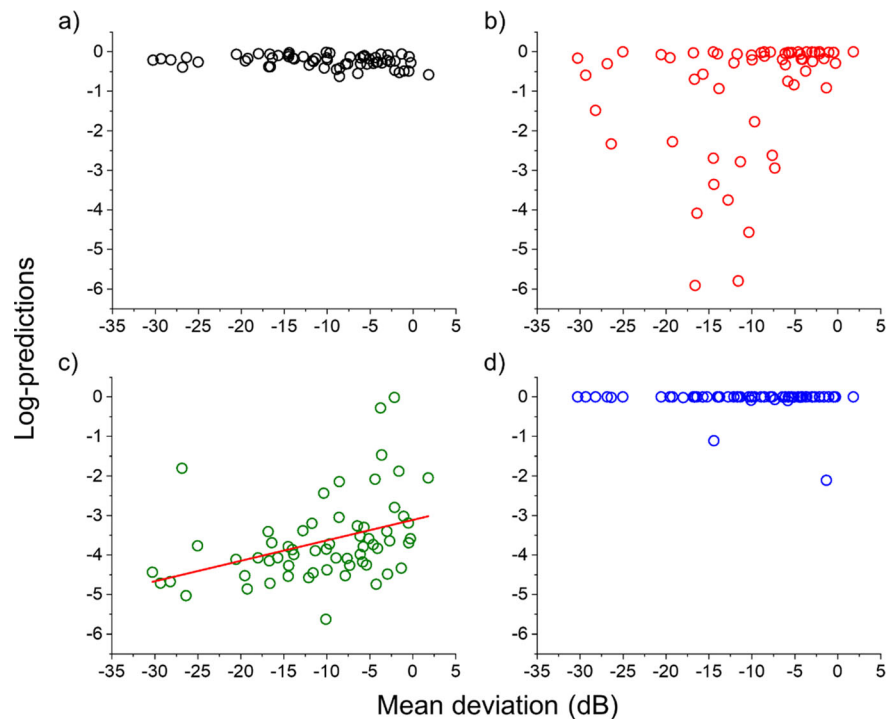
Fig. 6 Visual explanations for feature maps in each CNN model. The heatmaps in **a–d** indicate the relative activation intensity for predicting glaucoma. These heatmaps are superimposed on the original color fundus photograph (**a:e**, **b:f**, **c:g**, **d:h**). The green model was localized to the optic disk and retinal nerve fiber layer with a high predictive value of 0.0019. The blue model was localized to the optic disk, but the predictive value was 0.914. A predictive value of less than 0.5 was considered to indicate glaucoma



focused on the optic disk and RNFL, which have typical patterns related to glaucoma (Fig. 6) [3, 4]. Nerve fiber layer defect is related with the visual field defects [27], and MD is one of the indicators to assess advanced visual field defects [27, 28]. The log-predictions of the green model for glaucoma were

significantly and positively correlated with those of MD (Fig. 7c). These findings suggest that a green contrast digital filter improves the diagnostic performance of the CNN model for glaucoma that reached the upper limit by tuning the parameters.

Fig. 7 Relationship between log-predictions and mean deviation. The black, red, green, and blue circles indicate the RGB (a adjusted $R^2 = 0.044$, $P = 0.057$), red (b adjusted $R^2 = 0.018$, $P = 0.151$), green (c adjusted $R^2 = 0.137$, $P = 0.002$), and blue (d adjusted $R^2 = 0.006$, $P = 0.42$) models, respectively. The red line indicates the regression line



The red and blue color channels have adverse effects on CNN using a small dataset (Figs. 3, 4). Red light is absorbed poorly by retinal pigment epithelium [25], and a choroidal pattern can be observed in fundus photographs. The choroidal pattern differs depending on the individual, and the contrast also depends on refractive errors [29]. We believe that the color information in the red channel contains artifacts that affect the construction of the CNN model, because the fundus image has a lot of red components due to blood vessels [23].

Blue light allows for better imaging of retinal layers, because it is absorbed by blood and by the retinal pigment epithelium [25]. Fundus photography in blue light provides a dark background on the top layers of the retina, such as RNFL. However, the color information for blue was insufficient to structure the CNN model using raw data from color fundus photographs in the present study. We set the brightness range between 0.2 and 1.0, although the baseline brightness did not increase in the present study. We consider this is a limitation of the study. The blue channel images increase the baseline brightness during preprocessing, improving the diagnostic performance for glaucoma.

The time per epoch was significantly shorter in the single-color models than in the RGB model in Experiment 3. The grayscale image has single channel of 8 bit compared with the color image that has three channels of 24 bit. We speculate that VGG16 requires three channels for input; however, the calculation is faster in the single-color models than in RGB model, because the monochrome image is padded with 0 except 1 channel.

The present study shows that the green digital filter is useful to structure CNN models for automatic discrimination of glaucoma using fundus photographs. Our findings suggest that preprocessing is an important step when creating a CNN model for the identification of a large number of retinal disease using color fundus photographs.

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Author contributions Substantial contributions are as follows: MH, AM contributed for the conception and design; KI for the acquisition of data; MH for analysis and interpretation of data and drafting the article or revising it critically for important intellectual content. MH, AM, TM, TH, JK, TS, KI contributed for the final approval of the version to be published, and MH,

AM, TM, TH, JK, TS, KI for the agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This investigation adheres to the tenets of the World Medical Association Declaration of Helsinki. The experimental protocols and consent procedures were approved by the Institutional Review Board of Teikyo University (approval no. 18-161).

Informed consent Informed consent was obtained from all subjects after explaining the nature and possible complications of the study.

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